HLYBRO S500 LOG REVIEW REPORT

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1 Task Overview

1.1 Objective

Analyzing flight logs is a critical step in any drone or UAV operation. We look at flight logs to see exactly what happened during a flight—catching sensor errors, tuning controls, spotting vibration or battery issues—so we can fix problems, fly more efficiently, and keep the drone safe and reliable. Some aspect that we can investigate include:

- Safety - Reliability
- Performance - Tuning
- Efficiency - Endurance
- Regulatory - Compliance
- Maintenance - Flight Time Management

In order to analyst we will take a look at different graphs presenting aspects of the drone. There are also data table and map but they are not really helpful in this situation because mostly they shown the constant parameters which we set initially for the drone. Graphs on the other hand, show real time data, which is useful in analysing the drone's behaviours in real time—things like sensor readings, motor commands, and GPS accuracy. By contrast, the plots reveal spikes, drifts, or delays that help us spot problems and fine-tune the drone's performance.

1.2 Basic information for log analyzing

1. Plot background color is used to indicate flight mode during recording (where graphs depend on mode):

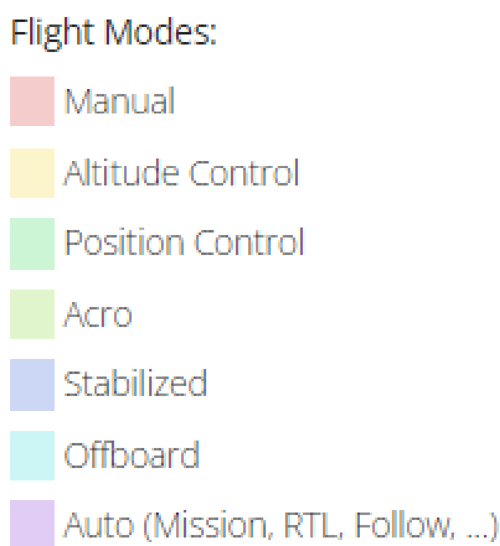

Figure 1: Color code for flight mode

2. Roll Pitch Yaw Analysis graphs often have 3 lines: Green line (Setpoint): This is the velocity/acceleration the autopilot is asking for. It's the command we or the mission send to the autopilot.

Red line (Estimated): This is the actual velocity/acceleration measured by the drone's sensors and filter. It shows what the aircraft is really doing.

Blue line (Digital Tuning): This is often the "I" term from the PID controller, i.e. the running correction the controller applies to eliminate any steady offset between the green and red lines.

2 General Information

Airframe:	Holybro S500 Quadrotor x (4015)	Distance:	185.3 m
Hardware:	PX4_FMU_V6X (V6X003)	Max Altitude Difference:	14 m
Software Version:	v1.14.3 (1dacb4cd)	Average Speed:	8.6 km/h
OS Version:	NuttX, v11.0.0	Max Speed:	18.1 km/h
Estimator:	EKF2	Max Speed Horizontal:	18.1 km/h
Logging Start ?:	04-07-2025 14:35	Max Speed Up:	5.4 km/h
Logging Duration:	0:01:17	Max Speed Down:	4.4 km/h
Vehicle Life Flight Time:	13 hours 43 minutes 16 seconds		
Vehicle UUID:	0006000000003931373131325102001d0022	Max Tilt Angle:	23.4 deg

Figure 2: General Information

Platform: A Holybro S500 X-frame running PX4 v1.14.3 on an FMU_V6X board with the EKF2 estimator. While EKF2 is a stable filter, it lacks many of the enhancements available in EKF3—such as support for beacon navigation, wheel encoders, and visual odometry.

(Source: ArduPilot: Extended Kalman Filter Overview)

Flight Duration: This mission is a short flight, only 1 min 17 s and covered only 185 m and climbed 14 m overall with an average speed of 8.6 km/h—suggesting a slow, cautious mission due to the testing environment limited to the testing zone, near the building and electric line.

Velocity: Peak horizontal speed was 18.1 km/h, vertical speeds were modest (5.4 km/h up, 4.4 km/h down), and the max tilt was 23.4°—all well within normal multirotor limits.

3 Analysis

3.1 Waypoint Mission

Figure 3: Mission Map

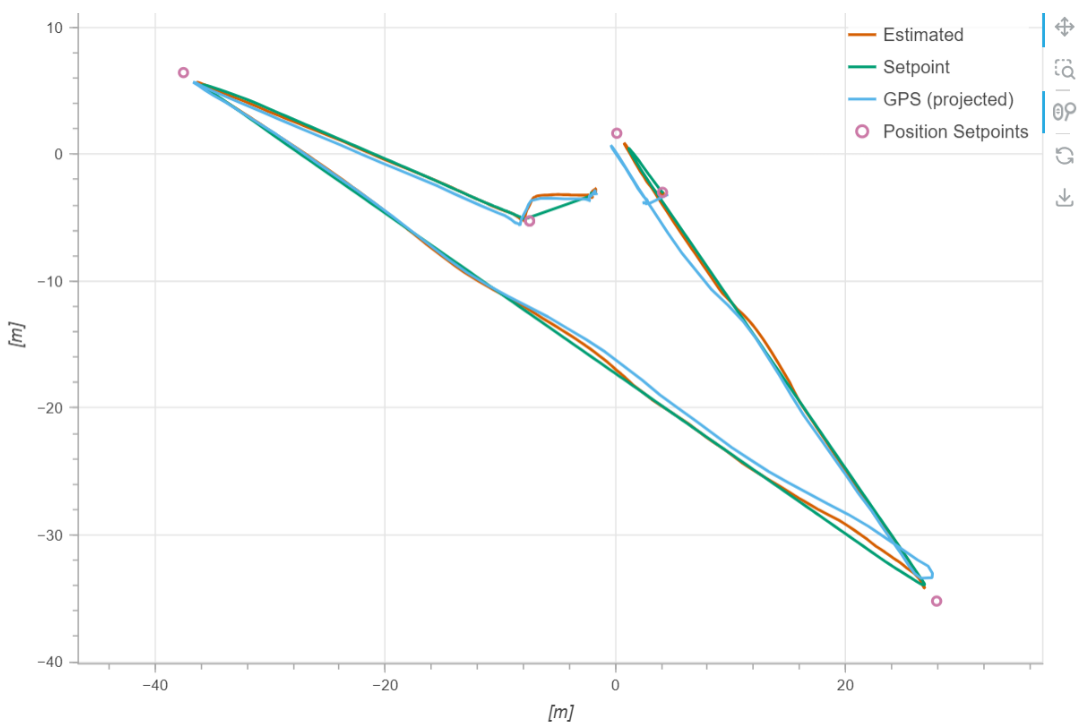

Figure 4: Flight path graph

Figure 4 shows the horizontal trajectory of the mission, plotting desired waypoints and path versus the drone's actual motion. The circles mark each programmed waypoint on QGControl, while the green line connects them to indicate the controller's ideal flight path (the setpoint). The orange trace is the EKF-filtered position estimate, and the light-blue line shows the raw GPS fixes projected into the same local frame. Notice how the estimated path closely follows the straight green segments, deviating by less than a meter even when the GPS data (blue) bounces around by a meter or two due to a lack of GPS signals. This tight tracking demonstrates that the vehicle's navigation and control loops are effectively smoothing out GPS noise and holding the aircraft on the course. The reason the path does not touch the waypoint entirely is that there is some tolerance radius at each waypoint, reducing the calculation workload. Overall, this is an accurate mission.

3.2 Altitude

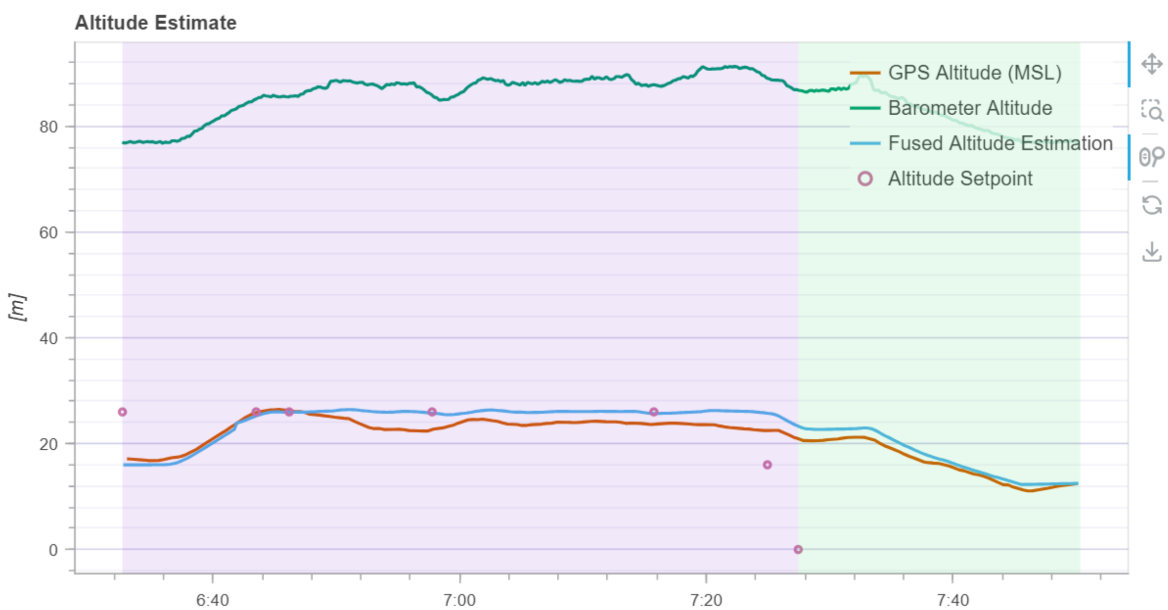

Figure 5: Altitude Estimation

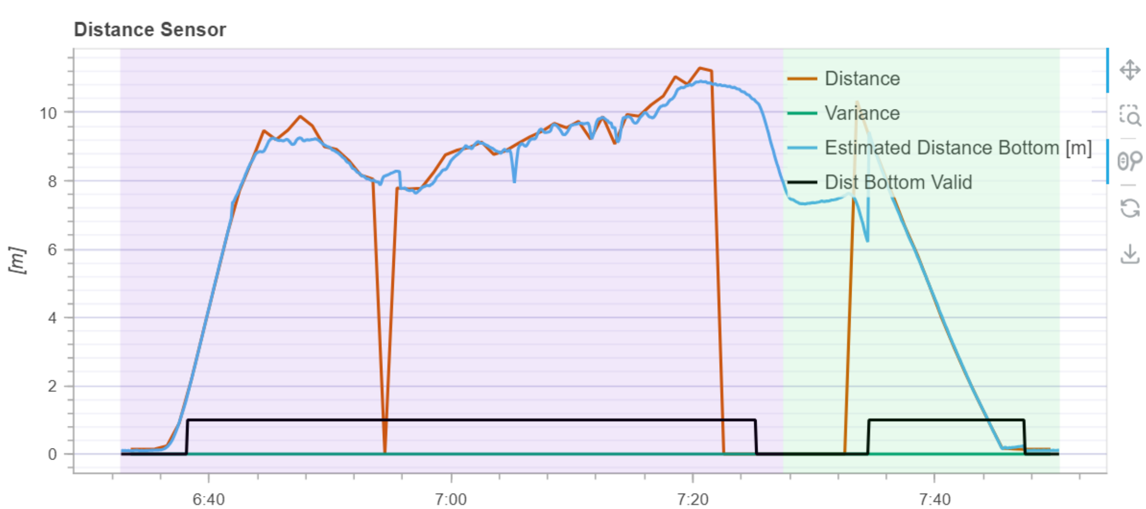

Figure 6: Distance sensor data

The mission commanded a 10 m climb, but the fused altitude trace sits between roughly 16 m and 26 m—an obvious offset due to barometric bias. Rather than fixate on that absolute error, we instead compare the minimum and maximum fused values, which differ by about 10 m, exactly matching the setpoint. The raw barometer readings span approximately 79 m to 89 m—again a 10 m range—confirming that, despite its bias, the baro reliably captures relative altitude changes.

For true altitude-above-ground measurements, however, we turn to the TF Mini Plus distance sensor (0–10 m range, ±1 m accuracy). In level terrain, this lidar-based sensor provides the most accurate real-time height data.

4 Roll - Pitch - Yaw

4.1 Roll

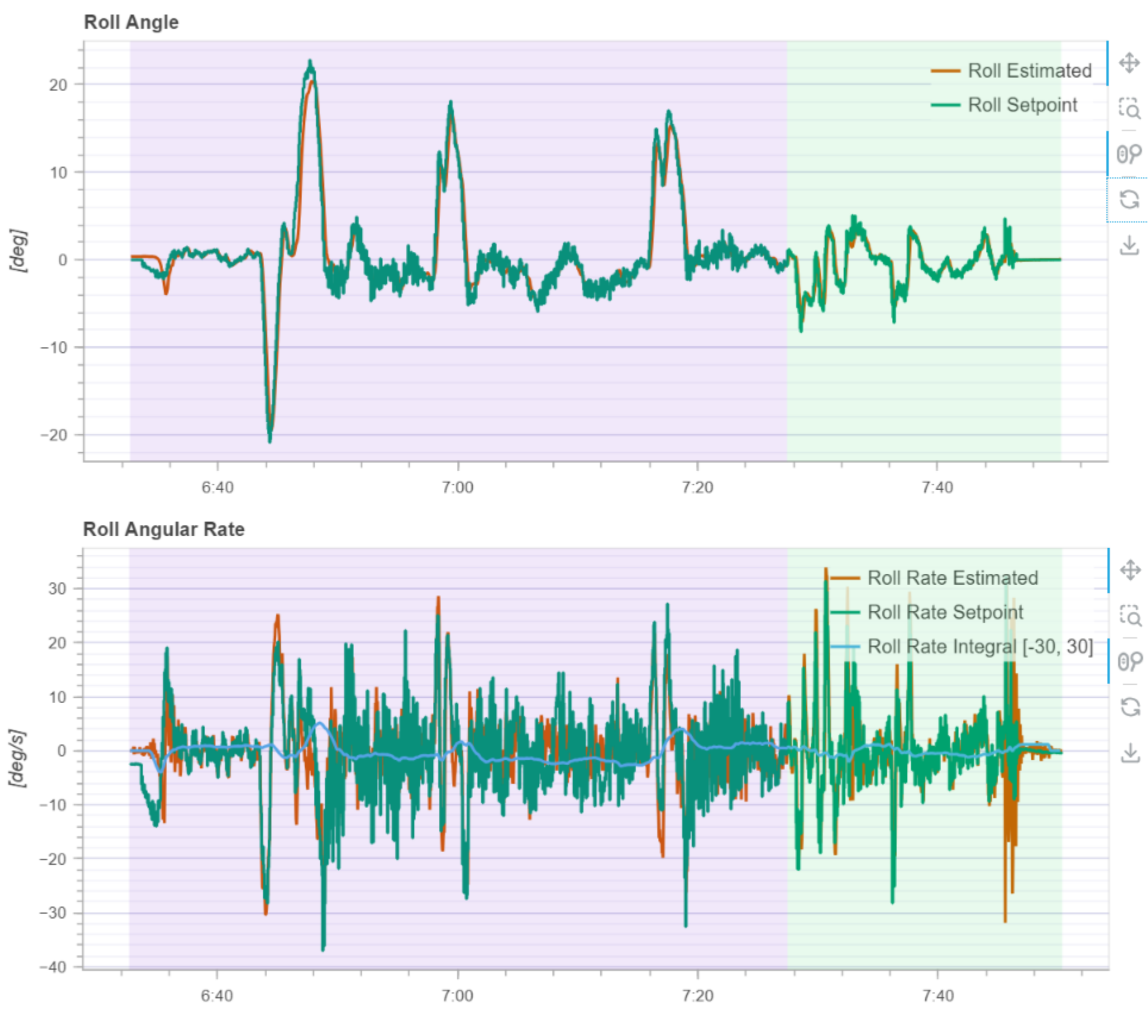

Figure 7: Roll Angle and Rate

Throughout the mission, the drone follows its roll setpoints—two ±20° turns early on, ±10° waypoint adjustments, and a near-zero angle during straight flight—with only a few degrees of lag, minimal overshoot, and fluctuations under ±5°, showing strong attitude control. Even after switching out of mission mode, rapid manual roll commands are tracked closely. The PID's fine-tuning (blue) line further smooths noise and bias to prevent overshoot, confirming effective stabilization against disturbances. The enormous fluctuation at the end presents the vibration of the landing process.

4.2 Pitch

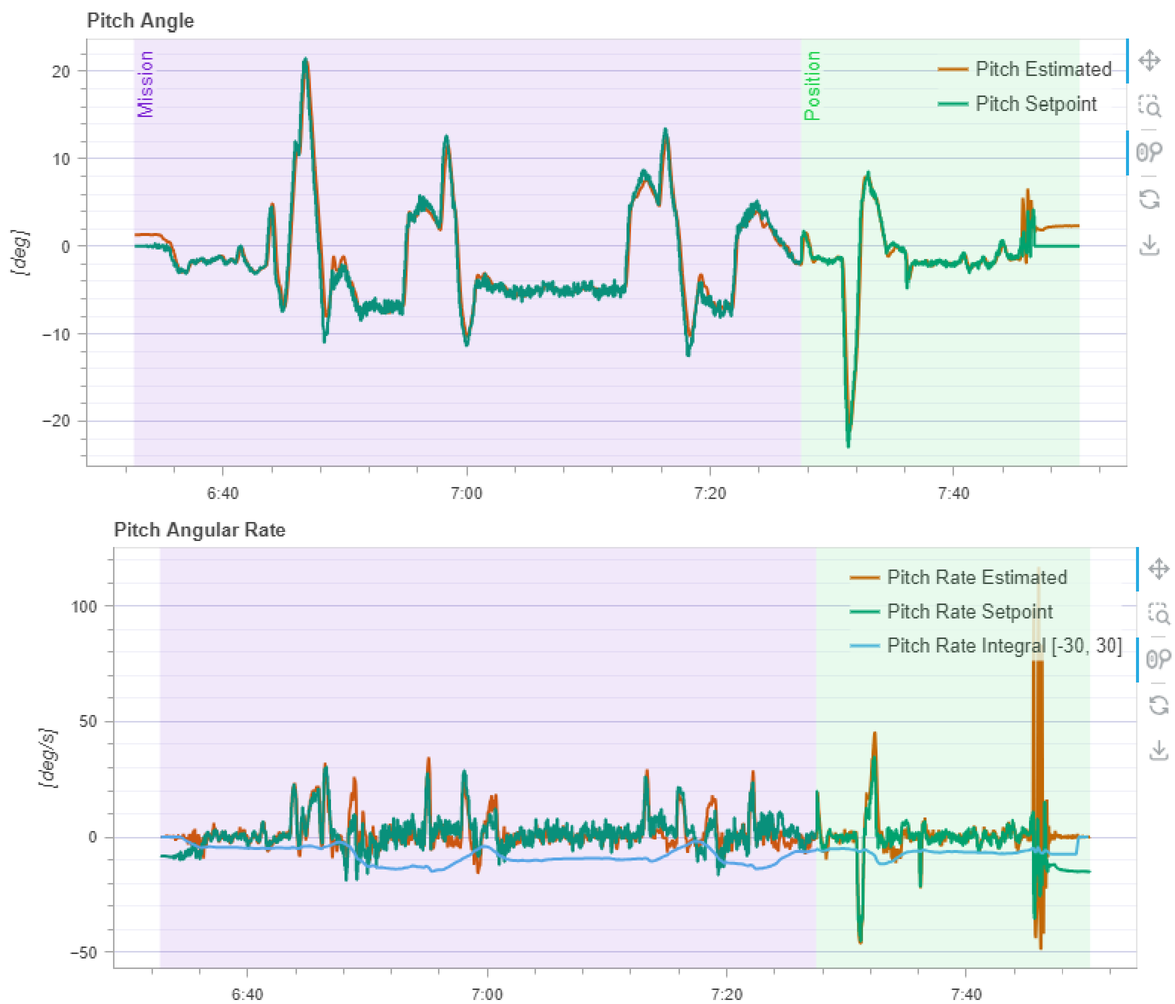

Figure 8: Pitch Angle and Rate

Overall, the pitch response remains well-behaved throughout the mission. The largest pitch excursions occur during waypoint turns, while the aircraft maintains a nearly level attitude on straight segments. In both angle and rate, the estimated signals track their setpoints with high fidelity—angle errors stay under 1° and rate errors under 5–7 °/s (below 10 % of commanded values). The PID's built-in filter further smooths the angular-rate signal, damping noise and preventing overshoot. Together, these results confirm robust and precise pitch control under both autonomous and manual inputs.

4.3 Yaw

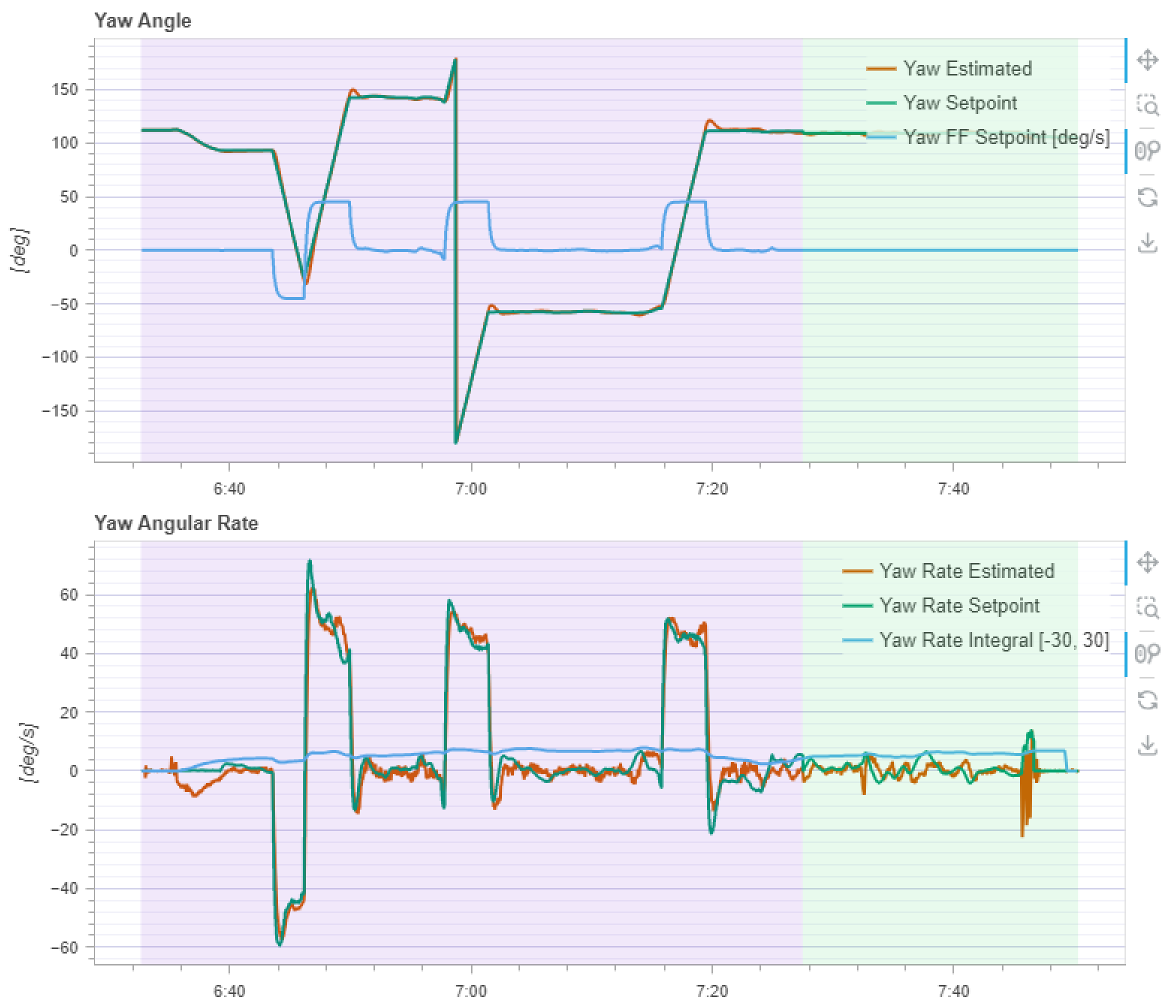

Figure 9: Yaw Angle and Rate

During each waypoint turn,the green line (the commanded yaw) jump sharply—sometimes by more than 150°—and the red line (the actual yaw) follows almost immediately, with only a 1–2° gap. This small, consistent difference shows the autopilot's rapid corrective action: whenever the setpoint shifts, the controller drives the aircraft to close that error within a fraction of a second. The blue trace is especially important here; it represents the PID's fine-tuning filter acting on the yaw-rate command. By continuously trimming out tiny biases and damping high-frequency noise, it prevents overshoot or oscillation, ensuring that the red line converges smoothly on the green one without any jittery swings.

There are sudden vertical strokes in the angle plot. Those aren't actual spins of the drone, but because when the setpoint jumps big gaps,e.g +150° to –170°, the plotting software connects those two points across the ±180° boundary with a straight line, producing a tall spike. Similarly, the feed-forward rate pulses (blue) and setpoint changes (green) happen almost instantaneously in the log, so connecting them creates what looks like a vertical line.

5 Local Position

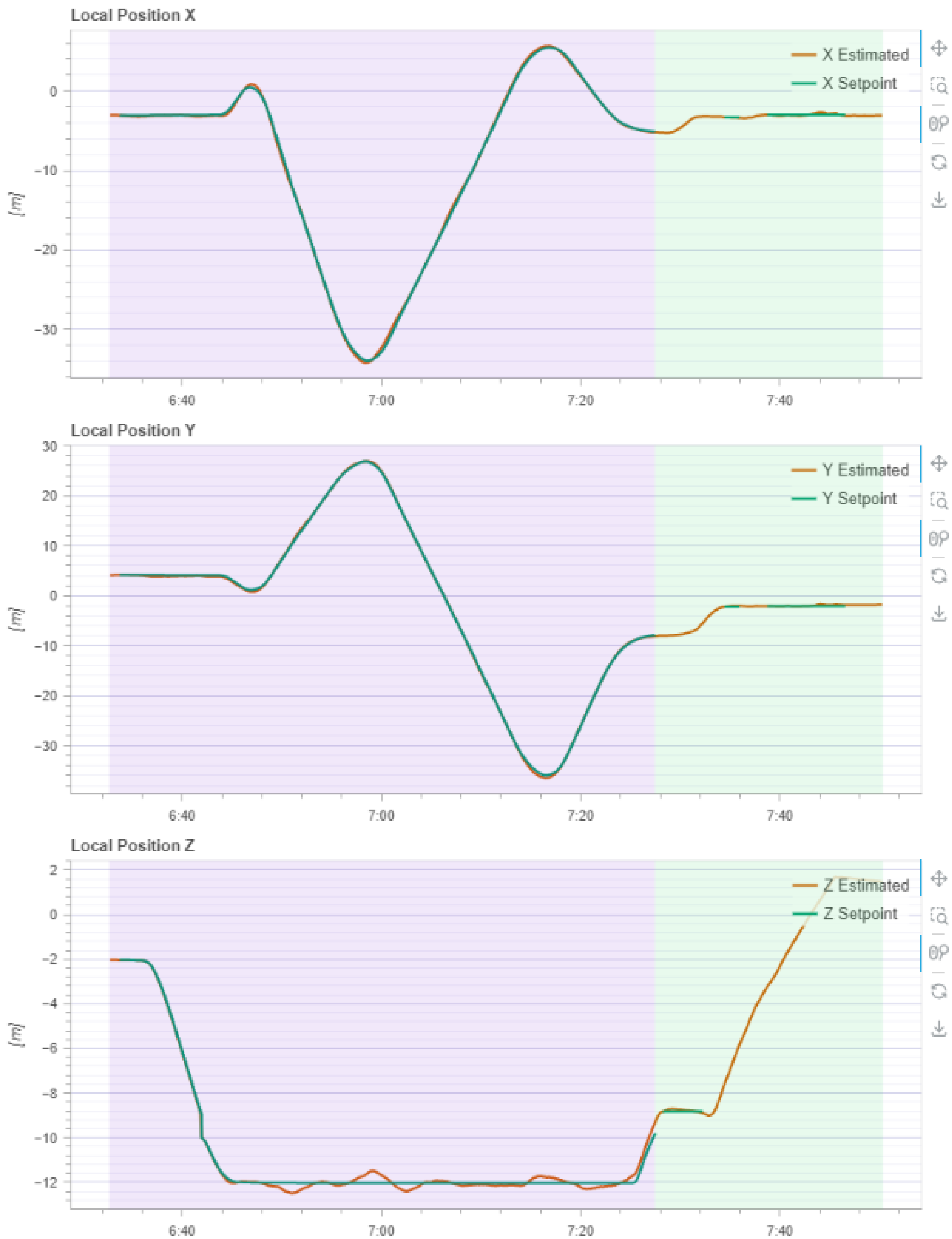

Figure 10: Local Position in X, Y, Z Axis

The UAV tracked its X and Y setpoints almost perfectly throughout the flight. In practice, small GPS drifts introduced slight offsets from the nominal path, but these were too subtle for the autopilot to detect—only a human operator comparing GPS to the planned waypoints would notice them. In the altitude plot, the Z-axis is inverted, yet it still clearly shows the vehicle climbing to the commanded 10 m. The small oscillations around each setpoint reflect the autopilot's active stabilization, as it continuously adjusts thrust and attitude to hold position and damp out disturbances.

5.1 Velocity in X, Y, Z Axis

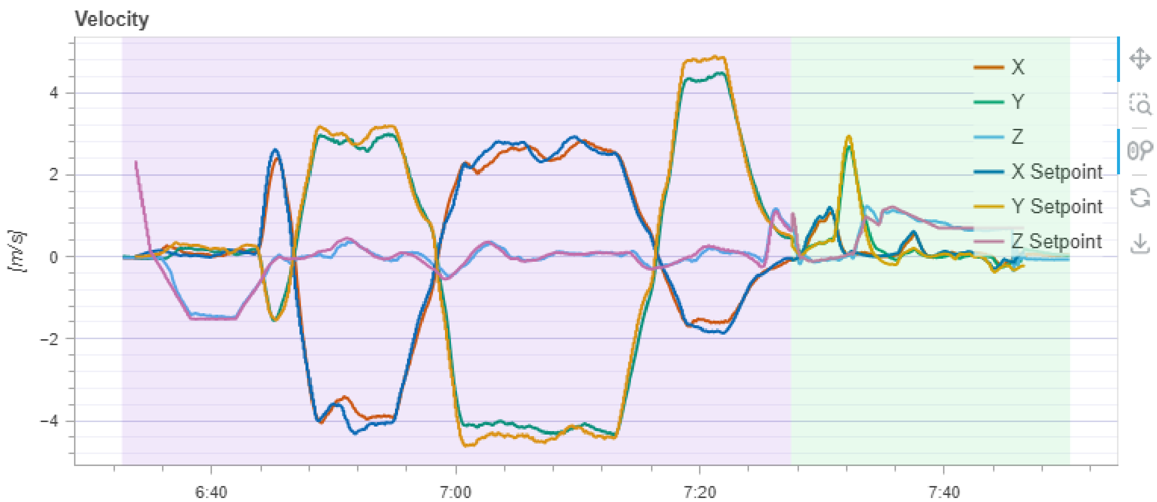

Figure 11: Velocity

The UAV's measured velocities in X (orange/yellow) and Y (green/teal) peak at ±5 m/s during each turn and settle back to zero almost instantly, with tracking errors under 0.2 m/s (< 4 %). The Z velocity (light-blue/purple) stays near zero in horizontal flight, with only brief ±1 m/s excursions on climbs and descents, rebounding within 0.1 m/s. Those tiny ±0.1–0.2 m/s ripples are the PID's damping of GPS noise and disturbances, showing highly accurate, stable velocity control across all axes.

5.2 Control

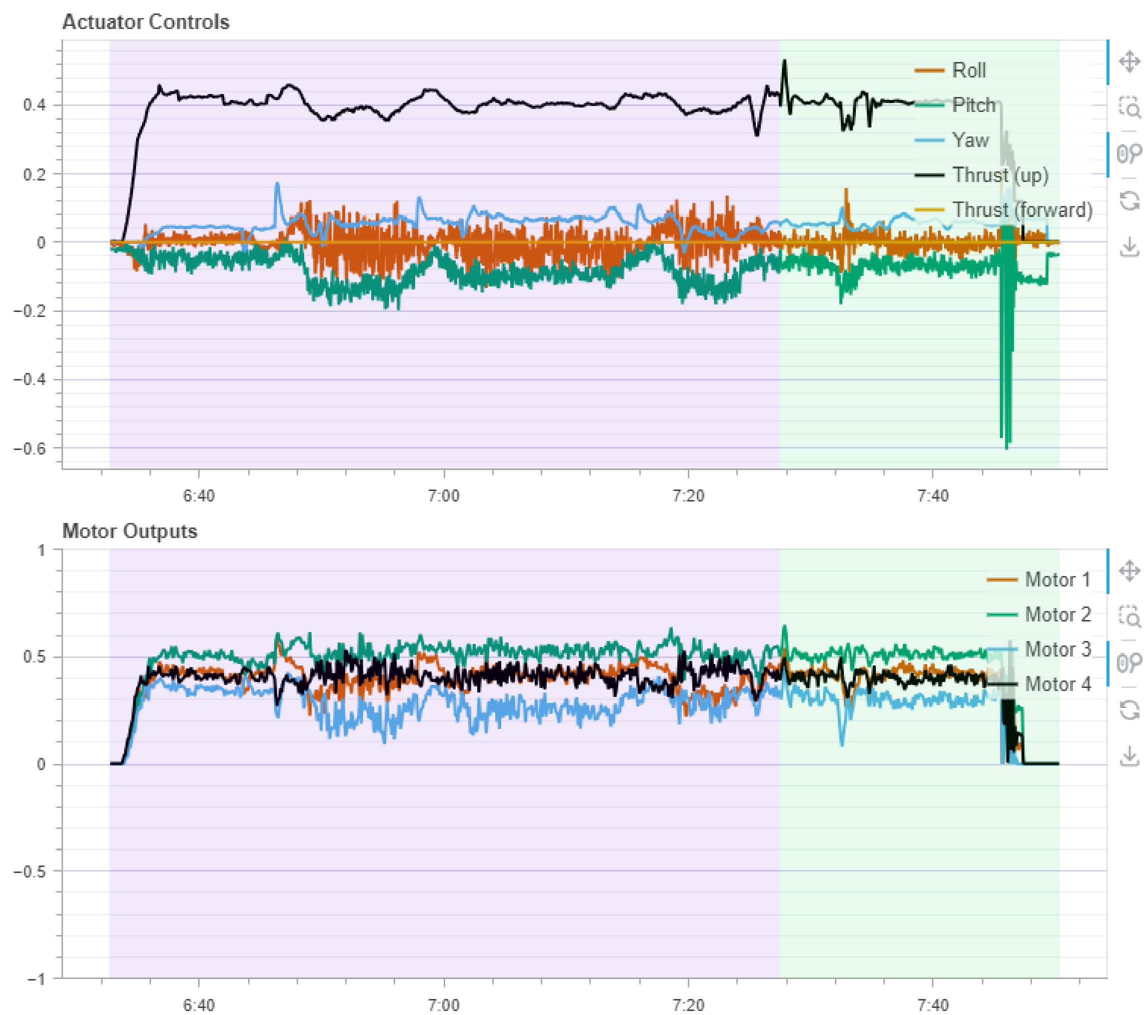

Figure 12: Control Actuators

The top plot shows the control signals sent to each axis and to thrust: the black line (up-thrust) quickly rises to about 0.4 – 0.45 at takeoff and stays there until descent, while the orange (roll), green (pitch) and blue (yaw) traces oscillate around zero—those small ripples are the autopilot nudging the craft to stay level and to turn when needed. The thin yellow line (forward thrust) barely moves, since most motion is horizontal or vertical.

In the lower plot, all four hover around 0.4–0.6 of maximum input, with tiny differences between them to generate roll, pitch, or yaw moments, which is normal due to the self-balancing function depending on the actuators' differences. At mission en,d the thrust and motor signals drop smoothly back to zero. Together, these plots show that the autopilot is issuing steady, well-balanced commands and that the motor mix accurately implements them.

6 Vibration

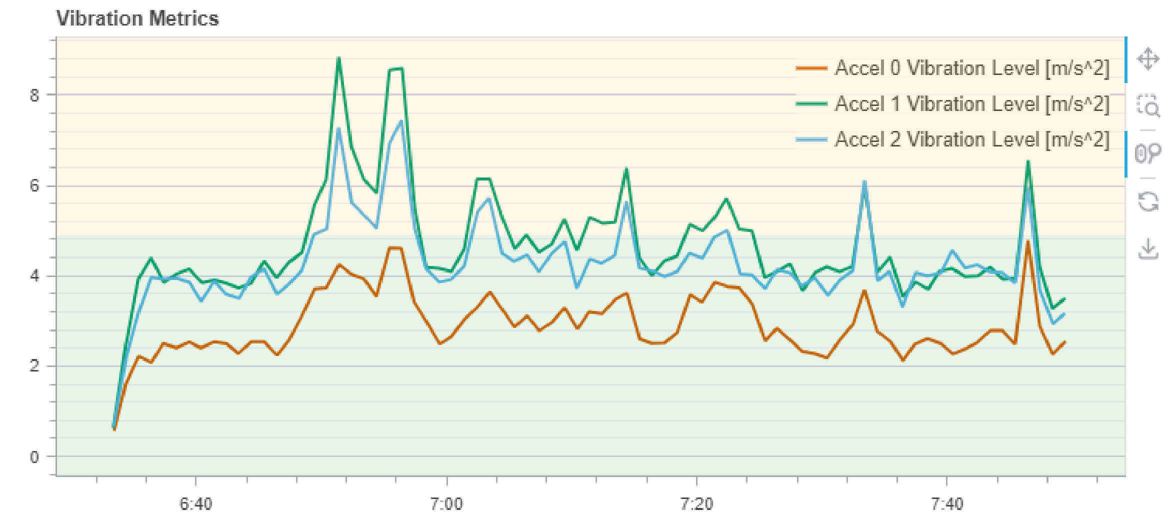

Figure 13: Vibration along axis

All three IMU axes show a quick rise in vibration immediately after takeoff, then settle into a fairly steady band for most of the mission before a final spike at landing. The lateral axis (Accel 1, green) and longitudinal axis (Accel 2, blue) hover around 4–6 m/s^2, while the vertical axis (Accel 0, orange) stays lower at about 2–4 m/s^2. You can see two prominent peaks—up near 9 m/s^2—during the sharpest turns around the waypoints; these correspond to high motor loads and airframe stress. Between those maneuvers, vibration levels drift only a little, indicating the autopilot and frame are damping most of the rotor-induced oscillations. Overall, the vibration remains within a typical operating envelope, so sensor data stay reliable and control loops aren't unduly disturbed.

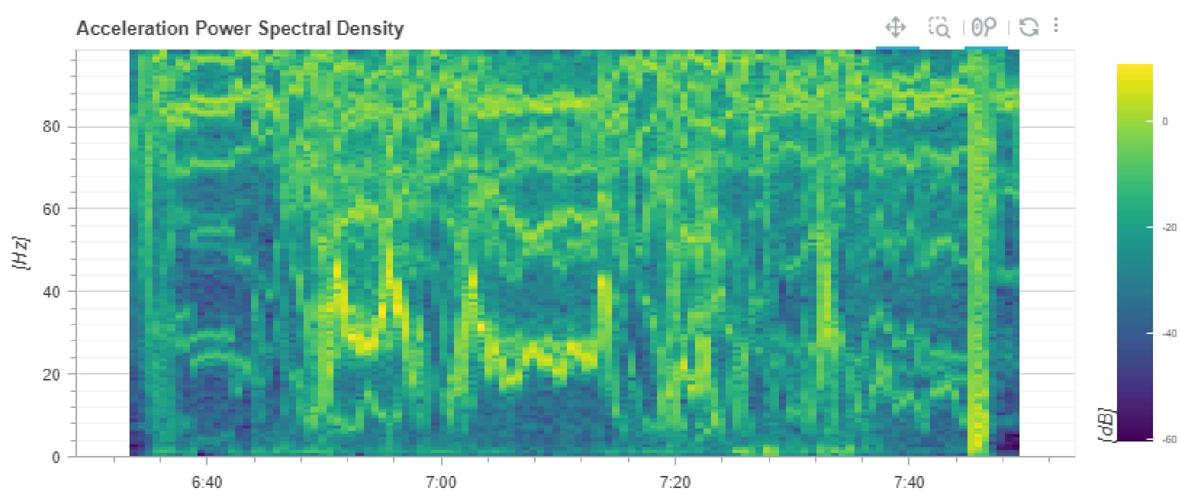

Figure 14: Acceleration Power Density

There is another way to analyst the vibration of the UAV, which is through the Spectral Density. From this figure, we not only see the vibration amount but also their frequency. In our log, the vibration is at an excellent amount of about 3%, which is represent by yellow-ich pixel on the spectrum. We can also see that the vibration frequency is around the range from 20 to

45 Hz. In this case, the vibration is at a lower frequency than the working frequency of the UAV (Typically 50Hz to 80Hz).

7 Power Managment

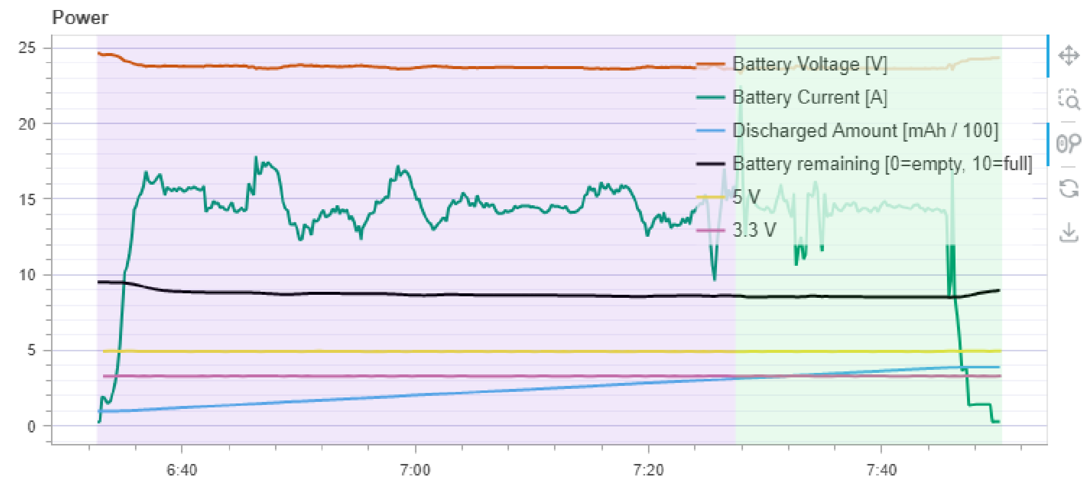

Figure 15: Power Graph

The power plot shows the battery performing very steadily throughout the flight. After takeoff, the voltage (orange) holds just under 24 V and only drops a little over time, which means the battery is healthy and not under too much load. The current draw (green) quickly rises to about 15 A when climbing, then varies between 12 A and 17 A during each sharp turn, and finally falls back toward zero at landing. The light-blue line tracks total discharged capacity, rising smoothly to about 4 Ah, while the black line shows remaining charge falling from full to around 80 %. Small dips in voltage and current when switching to position-hold reflect the autopilot fine-tuning the throttle. Overall, the battery delivered power smoothly, stayed in a good voltage range, and used only about 20

8 PID

Proper PID tuning is vital because it balances responsiveness and stability in the control loop:

- Proportional (P) gives an immediate correction to any error, but if too high it causes overshoot; if too low, the system responds sluggishly.
- Integral (I) removes steady-state error by accumulating past deviations, but if over-emphasized it can wind up and lead to slow oscillations.
- Derivative (D) predicts future error trends to dampen oscillations, yet if set too aggressively it amplifies sensor noise.

When these three gains are tuned correctly, the controller tracks setpoints quickly without overshooting, smooths out disturbances, and resists jitter. In a UAV, that means precise attitude and altitude control, efficient power use, and safer, more predictable flight performance.

8.1 PID of Angular Rate

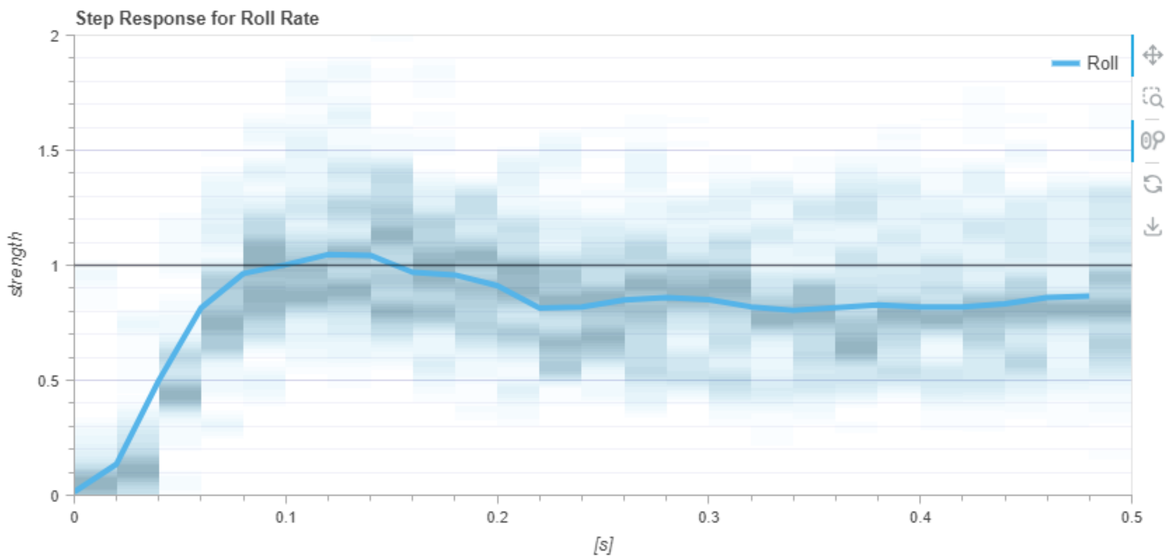

Figure 16: Roll Rate PID

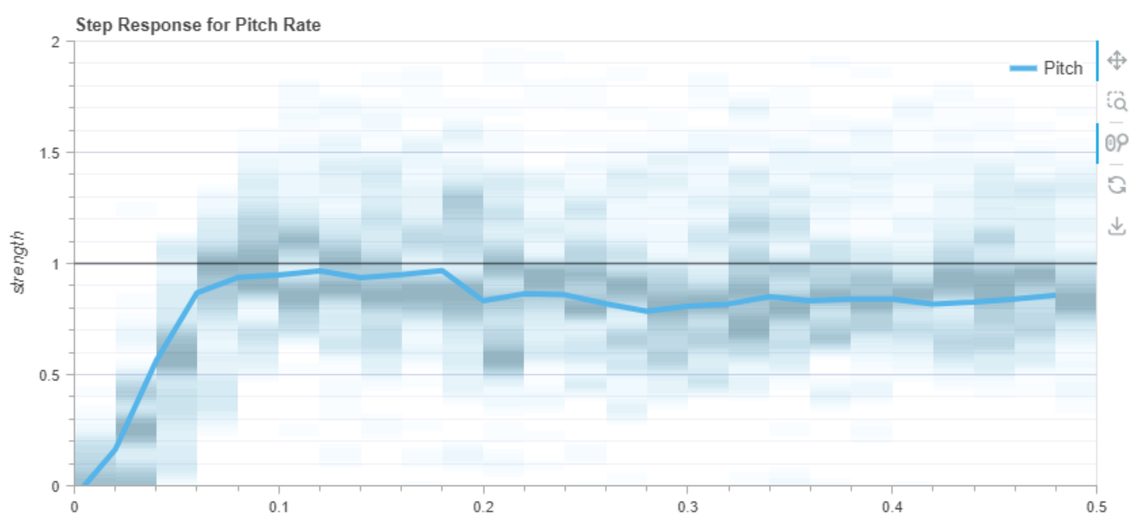

Figure 17: Pitch Rate PID

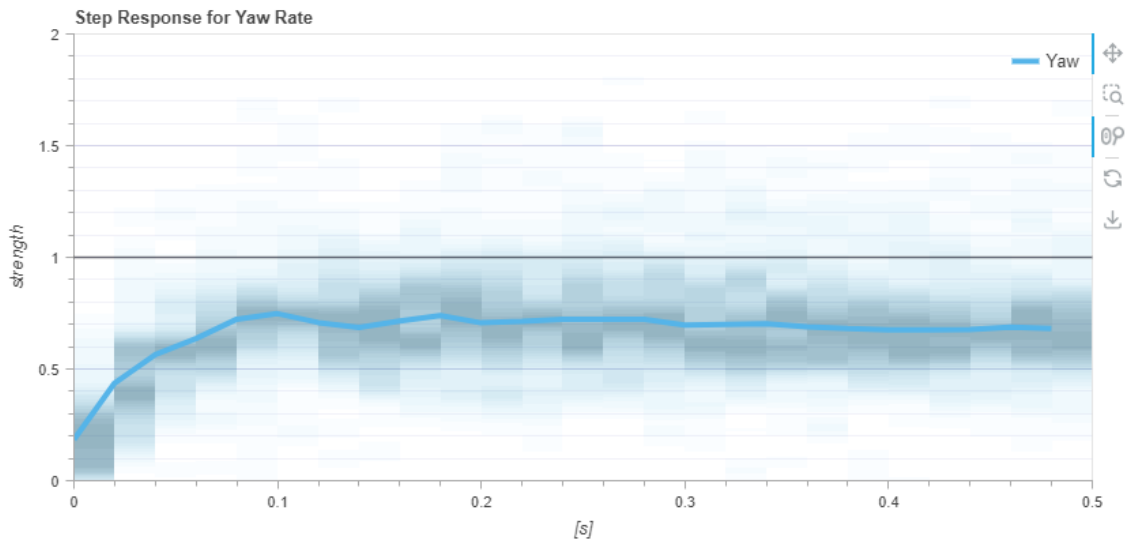

Figure 18: Yaw Rate PID

When looking at this graph, we would want the line to follow the value 1 horizontally as close as possible. So in in the Roll rate, we have some overshoot, but the dropped bellow 1 and cannot go back after that. In other 2 graph (Pitch and Yaw rate), the line don't even reach 1, with a difference of 0.05 for Pitch and $>$ 0.2 for Yaw. This might indicate that the P element in the tuning process is set bellow the nessescary amount.

Roll:Is the fastest: it slightly overshoots (2 %) at 0.08 s, then undershoots and finally settles near unity around 0.3 s. Its very narrow heatmap reveals excellent consistency, since roll has the smallest moment of inertia and can tolerate more aggressive gains.

Pitch: Responds more quickly, climbing to roughly 0.95 of the target in 0.1 s, with a small overshoot (5 %) and settling by 0.2 s. The tighter heatmap shows more repeatable behavior than yaw, thanks to higher pitch control authority.

Yaw: Rises more slowly than the others, reaching only about 0.75 of the command by 0.1 s and then very gradually drifting toward the setpoint with no overshoot. The broad, diffuse heatmap indicates moderate trial-to-trial variability, reflecting yaw's larger rotational inertia and conservative tuning to avoid oscillation

By increasing the P-term, the controller delivers a stronger immediate correction, shortening the rise-time toward unity. A higher D-term then damps any overshoot or oscillation caused by that more aggressive P, keeping the response clean. Finally, a small boost to the I-term pulls out any remaining steady-state error so the output settles exactly on 1.0 and the loop can react to faster changes.